

VEKTOLUX VEHICLE ESCROW & SETTLEMENT ENGINE

Production Technical Architecture, PostgreSQL Relational Schema, Financial Ledger & Telco Integration

Operating Territory: Sierra Leone (SLE Currency) • **Telco Rails:** Orange Money & Africell Afrimoney • **Regulatory Standards:** BSL Guidelines & SLRSA Protocol
Publication Date: September 2026 • **Classification:** Enterprise Architecture Specification • **Version:** 1.0 Production

1. Executive Architecture & System Topology

The Vektolux Vehicle Escrow Engine provides cryptographic, milestone-driven, and legally auditable fund segregation for commercial vehicle rentals and sales across Sierra Leone. Engineered to meet Bank of Sierra Leone (BSL) Mobile Money Guidelines and SLRSA statutory requirements, it guarantees balance isolation, immutable double-entry ledgering, and zero double-spending.

Core Platform Invariants:

- **Balance Isolation:** Available (withdrawable) funds are stored separately from Escrow (locked) funds in all database transactions.
- **Mathematical Double-Entry:** Every transaction generates matching debits and credits: $\sum \text{Debit} - \sum \text{Credit} \equiv 0.00$.
- **Concurrency & Idempotency:** Handled via PostgreSQL pessimistic row locks (`SELECT ... FOR UPDATE`) and cryptographic webhook signatures.

2. Financial Split, Commission & Fee Computation Rules

All financial values are denominated in **Sierra Leone New Leones (SLE)** stored as `NUMERIC(14, 2)`.

2.1 Vehicle Rental Financial Model (60/40 Milestone Split)

- **Upfront Inflow into Escrow:** Base Rental Charge + Refundable Damage Deposit.
- **Platform Commission:** 15% levied *strictly on the Base Rental Charge* (0% on deposits).
- **Milestone 1 (Handoff / Trip Start):** 60% of Net Owner Proceeds disbursed to Owner upon pre-trip inspection approval; 60% of platform fee recognized as revenue.
- **Milestone 2 (Vehicle Return):** Remaining 40% disbursed to Owner upon return; 100% of damage deposit automatically refunded to Renter (or repair costs deducted and remitted to owner).

Milestone / Component	Mathematical Formula	5-Day Rental Example (SLE)
Base Rental Charge	Daily Rate × Days (SLE 1,500 × 5)	SLE 7,500.00

Milestone / Component	Mathematical Formula	5-Day Rental Example (SLE)
Refundable Damage Deposit	Fixed security deposit	SLE 2,500.00
Total Upfront Escrow Inflow	Base Rental + Damage Deposit	SLE 10,000.00
Platform Commission (15%)	$0.15 \times \text{Base Rental Charge}$	SLE 1,125.00
Net Owner Rental Proceeds	$0.85 \times \text{Base Rental Charge}$	SLE 6,375.00
Milestone 1: Handoff Payout (60%)	$0.60 \times \text{Net Owner Rental}$	SLE 3,825.00 (Disbursed to Owner Available)
Milestone 2: Return Payout (40%)	$0.40 \times \text{Net Owner Rental}$	SLE 2,550.00 (Disbursed to Owner Available)
Deposit Refund to Renter	100% of Damage Deposit	SLE 2,500.00 (Refunded to Renter Available/Mobile Money)

2.2 Vehicle Purchase / Sale Financial Model

1. **Earnest Inspection Fee:** Buyer deposits earnest fee (e.g. SLE 500.00) into escrow to lock the vehicle.
2. **Physical Mechanic Inspection:** Certified mechanic scans dual QR codes and inspects vehicle condition.
3. **Full Balance Locked:** Buyer funds remaining vehicle balance into escrow.
4. **SLRSA Ownership Transfer:** Official SLRSA Form C and endorsed Vehicle Registration Blue Book uploaded and audited. Platform deducts commission and remits net sale proceeds to seller.

3. Finite State Machine (FSM) Lifecycle

State	Trigger / Event	Financial Balance State
INITIATED	Order created; invoice generated	Zero funds committed
PENDING_PAYMENT	Telco USSD prompt pushed to mobile	Awaiting webhook confirmation
HELD_IN_ESCROW	100% upfront funds confirmed by telco	Funds locked in Renter Escrow Balance
PARTIALLY_RELEASED	Pre-trip inspection passed & QR scanned	60% base rental credited to Owner Available Balance
POST_INSPECTION_PENDING	Vehicle returned at dropoff location	Remaining 40% + 100% deposit held
SETTLED	Post-trip inspection clean & sign-off	40% to Owner; 100% deposit refunded to Renter
DISPUTED	Damage reported or odometer discrepancy	Deposit frozen pending admin repair quote
REFUNDED	Order cancelled prior to handoff	100% funds refunded to Renter

4. Production PostgreSQL Relational Schema

```
-- 1. WALLETS (Available vs Escrow-Locked Separation)
CREATE TABLE wallets (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  user_id UUID NOT NULL UNIQUE,
  currency VARCHAR(3) NOT NULL DEFAULT 'SLE' CHECK (currency = 'SLE'),
  available_balance NUMERIC(14, 2) NOT NULL DEFAULT 0.00 CHECK (available_balance >= 0.00),
  escrow_balance NUMERIC(14, 2) NOT NULL DEFAULT 0.00 CHECK (escrow_balance >= 0.00),
  total_lifetime_earnings NUMERIC(14, 2) NOT NULL DEFAULT 0.00,
  version BIGINT NOT NULL DEFAULT 1,
  created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
  updated_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);

-- 2. ESCROW ORDERS
CREATE TABLE escrow_orders (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  order_code VARCHAR(32) NOT NULL UNIQUE,
  order_type VARCHAR(32) NOT NULL, -- VEHICLE_RENTAL, VEHICLE_PURCHASE
  renter_or_buyer_id UUID NOT NULL,
  owner_or_seller_id UUID NOT NULL,
  vehicle_listing_id VARCHAR(64) NOT NULL,
  currency VARCHAR(3) NOT NULL DEFAULT 'SLE',
  base_rental_amount NUMERIC(14, 2) NOT NULL DEFAULT 0.00,
  refundable_deposit_amount NUMERIC(14, 2) NOT NULL DEFAULT 0.00,
  gross_escrow_amount NUMERIC(14, 2) NOT NULL,
  platform_fee_amount NUMERIC(14, 2) NOT NULL DEFAULT 0.00,
  split_60_released_amount NUMERIC(14, 2) NOT NULL DEFAULT 0.00,
  split_40_released_amount NUMERIC(14, 2) NOT NULL DEFAULT 0.00,
  deposit_refunded_amount NUMERIC(14, 2) NOT NULL DEFAULT 0.00,
  status VARCHAR(32) NOT NULL DEFAULT 'INITIATED',
  created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
  updated_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);

-- 3. DOUBLE-ENTRY LEDGER (Immutable Balancing)
CREATE TABLE ledger_transactions (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  transaction_code VARCHAR(64) NOT NULL UNIQUE,
  escrow_order_id UUID REFERENCES escrow_orders(id),
  description TEXT NOT NULL,
  created_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);

CREATE TABLE ledger_entries (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  transaction_id UUID NOT NULL REFERENCES ledger_transactions(id),
  account_type VARCHAR(32) NOT NULL,
  user_id UUID,
  direction VARCHAR(6) NOT NULL, -- DEBIT, CREDIT
  amount NUMERIC(14, 2) NOT NULL CHECK (amount > 0.00),
  currency VARCHAR(3) NOT NULL DEFAULT 'SLE'
);
```

5. End-to-End REST API Specifications

1. POST /api/v1/vehicles/escrow/initiate

Initiates an escrow agreement, calculates the 15% platform commission, sets up the 60/40 milestone schedule, and triggers the mobile money prompt.

```
// Request Payload
{
  "orderType": "VEHICLE_RENTAL",
  "vehicleListingId": "veh_tipper_truck_501",
  "rentalPeriod": { "startDate": "2026-09-20T08:00:00Z", "endDate": "2026-09-25T18:00:00Z", "numberOfDays": 5 },
  "pricing": { "dailyRateSLE": 1500.00, "baseRentalAmountSLE": 7500.00, "refundableDepositAmountSLE": 2500.00 },
  "paymentMethod": { "provider": "ORANGE_MONEY_SL", "subscriberMsisdn": "23276123456" }
}

// Response (201 Created)
{
  "success": true,
  "data": {
    "escrowOrderId": "b78b87d2-4328-4e61-a08b-b152d1197ef4",
    "orderCode": "VK-ESC-2026-88190",
    "status": "PENDING_PAYMENT",
    "financialSummary": {
      "baseRentalAmount": 7500.00,
      "refundableDepositAmount": 2500.00,
      "totalUpfrontEscrow": 10000.00,
      "platformFeeAmount": 1125.00,
      "netOwnerProceeds": 6375.00,
      "milestonePayouts": { "handoff60Percent": 3825.00, "return40Percent": 2550.00, "depositRefundOnReturn": 2500.00 }
    }
  }
}
```

2. POST /api/v1/vehicles/inspection/complete

Submits 6-angle directional vehicle photos, odometer reading, fuel %, and verifies physical presence via dual QR code scan.

3. POST /api/v1/vehicles/escrow/release-milestone

Executes the 60% handoff milestone release to the vehicle owner upon pre-trip inspection approval.

4. POST /api/v1/vehicles/escrow/settle-return

Finalizes the rental: disburses the remaining 40% to the owner and refunds 100% of the damage deposit to the renter (or deducts repair costs if damage was assessed).

6. Telco Webhook Integration & Double-Spending Prevention

Webhooks from Orange Money and Africell Afrimoney are processed inside serial transactions with the following safeguards:

- **Cryptographic Verification:** HMAC-SHA256 signature verification over (transactionId | orderCode | amount | status).
- **Deduplication Guard:** telco_webhook_logs maintains a unique index on (provider, external_transaction_id). Replay callbacks return HTTP 200 immediately without re-crediting.
- **Pessimistic Locking:** Wallets are locked using SELECT ... FOR UPDATE in sorted user ID order to eliminate deadlock risks.
- **Double-Entry Ledger Balancing:** All balance updates require paired DEBIT and CREDIT ledger entries ensuring mathematical parity.